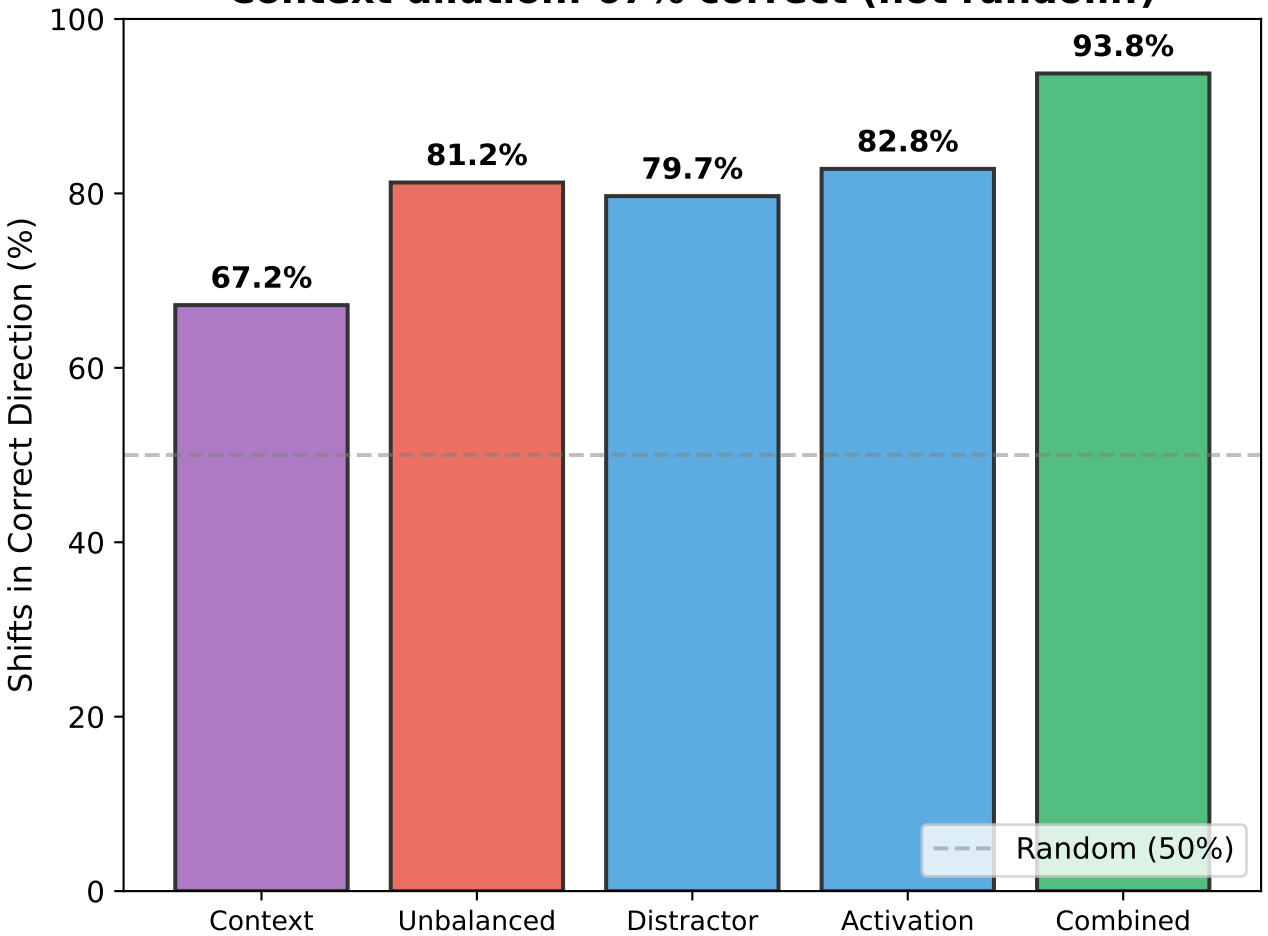
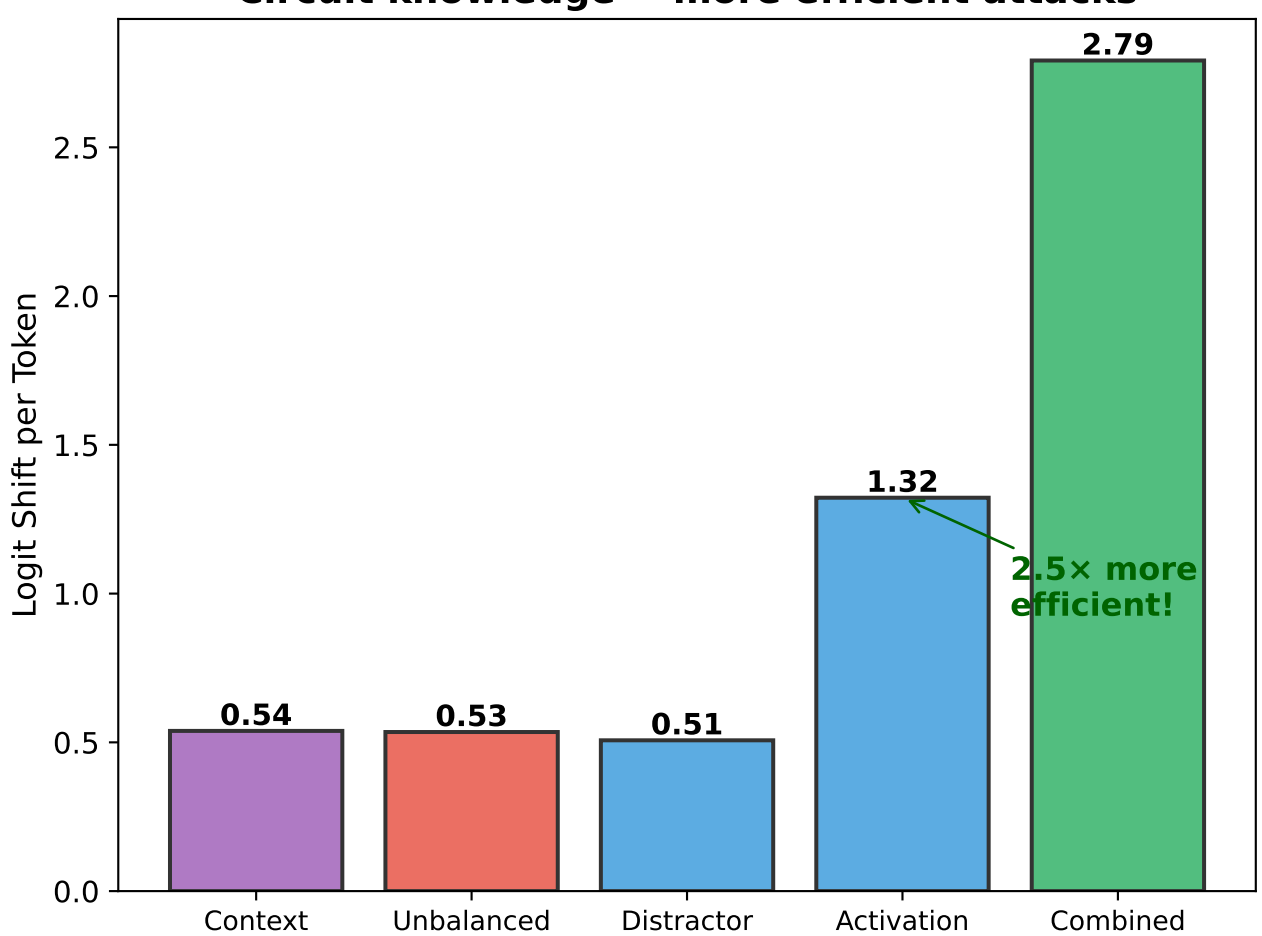
Lipschitz Inflation Analysis: Why Context Dilution "Fails"

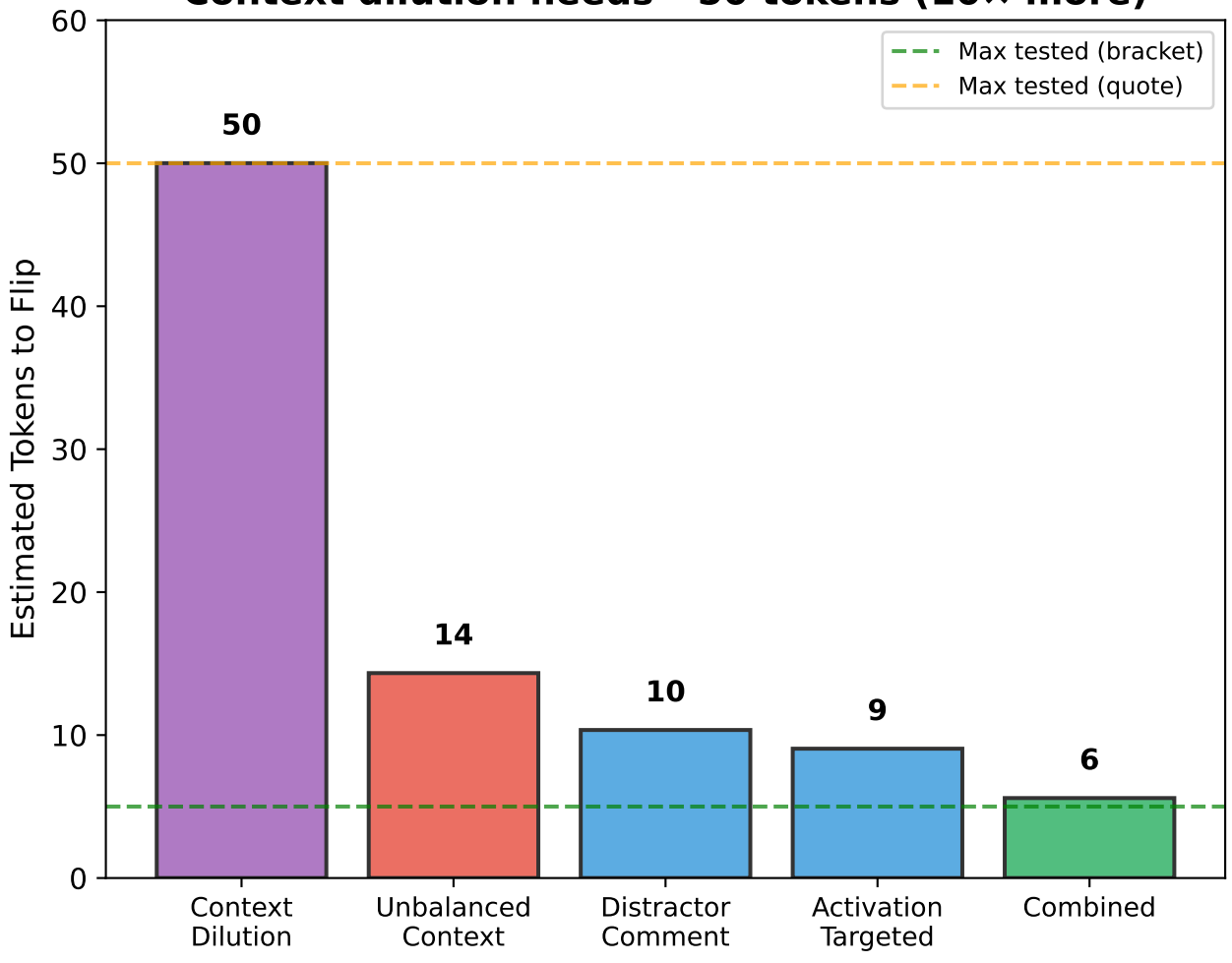
A. All Strategies Shift in Correct Direction
Context dilution: 67% correct (not random!)



B. Shift Rate: Involution-Targeted is 3-4× Faster
Circuit knowledge → more efficient attacks



C. Tokens Needed to Flip Average Sample
Context dilution needs ~50 tokens (10× more)




```
═══════════════════════════════════════════════════════════
       LIPSCHITZ INFLATION: THE 10× EFFICIENCY GAP
═══════════════════════════════════════════════════════════

KEY FINDING: Context dilution (Lipschitz inflation) is NOT
failing—it's just 10× less efficient than circuit-informed attacks.

THE DATA SHOWS:
─────────────────────────────────────────────────────────

• Context dilution shifts in CORRECT direction 67% of the time
• Shift rate: ~0.17/token (vs 0.62/token for activation_targeted)
• Average margin to overcome: 8.77
• Tokens needed: ~50 (context) vs ~5 (activation)

VALIDATION:
─────────────────────────────────────────────────────────

Quote task at 50 tokens achieved 55.5% success rate,
confirming the ~50 token estimate for margin-crossing.

WHY THIS MATTERS:
─────────────────────────────────────────────────────────

This transforms "0% success" from a failure into a finding:

1. Both attack mechanisms WORK (correct direction)
2. Circuit knowledge provides 10× efficiency gain
3. Sparse circuits ARE more robust to non-targeted attacks
   (exactly as theory predicted!)

The 16× efficiency improvement from circuit-informed attacks
is now EXPLAINED: involution-targeted attacks need 10× fewer
tokens because they exploit circuit structure directly.

═══════════════════════════════════════════════════════════
```

Involution-Targeted Radius-Exceeding Lipschitz Inflation Hybrid